

Urban mobility analysis: Optimizing the flow of public traffic through Big data processing

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Abstract—The abstract goes here.

1. Introduction

Modern public transit system such as RED and Metro in Santiago generate a continuous massive stream of transactional data. Every passenger interaction validation a card at a turnstile produces a record characterized by high density. These smart card validation systems act as the primary sensor network for urban mobility, capturing millions of daily events that reflect the dynamic movement of people across the metropolitan area.

Despite the wealth of data generated by urban transit, transport authorities face major operational and computational challenges. First, long travel times and suboptimal route configurations severely erode the quality of life for daily citizens, understanding these extensive displacement patterns is critical for system improvement. Also high passenger congestion dynamically collapses specific stations and stops, especially during peak hours, the lack of rapid, scalable data analysis prevents real-time reallocation to alleviate these critical bottlenecks. Furthermore traditional Relational Database Management System are computationally ill-equipped to process and query gigabytes of high-density data efficiently [1], processing millions of daily transactional records requires transitioning toward distributed Big Data architectures (such as Batch and Stream Processing ecosystems) capable of handling volume and high generation velocity [1].

Transitioning from traditional analytical approaches to Big Data ecosystems allows transport engineering to shift from reactive decision-making-based on static monthly reports to proactive and predictive management [1].

By processing massive historical validation records alongside spatial transit models, it becomes possible to accurately infer Origin-Destination matrices and optimize vehicle operational frequencies. This optimization reduces operational expenditures for service providers while directly minimizing daily travel times for citizens, returning valuable time to commuters and delivering a clear social impact.

2. Datasets

Acá debe mencionar, referenciar, y describir los datasets con los que desea trabajar su problema.

2.1. Smart Card Validation Records (DTPM)

The primary transactional source will be the historical validation records provided by the Directorio de Transporte Público Metropolitano (DTPM) [1]. This dataset contains massive, anonymized records of passenger transactions upon entering the system (RED).

- **Structure and key variables:** The information is presented in heavy tabular formats and includes the exact timestamp of the transaction, an anonymized card identifier, and the ID of the boarding stop or station.
- **Data limitation and technical justification:** The system operates under a tap-in only logic, meaning the passenger's actual destination is a nonexistent data point in the raw database. To accurately map congestion, it is necessary to reconstruct trips by applying Alighting Estimation algorithms. This requires processing the total daily volume by grouping records by card ID, sorting them chronologically, and calculating the destination by inferring spatial proximity to the user's next trip. This massive operation of sorting, crossing, and sequential inference over millions of rows overwhelms traditional relational databases, making the use of Batch Processing architectures strictly necessary.

2.2. General Transit Feed Specification (GTFS) Data

To provide spatial context to the transactional data and enable the destination inference algorithms, information from the GTFS standard—obtained from the DTPM and the State's Open Data Portal [2] will be integrated.

- **Structure and key variables:** It consists of inter-related text files that provide the exact geographic

coordinates (latitude and longitude) of each stop and station, the spatial shape of the routes, and the scheduled transit times.

- **Relevance:** When a user makes their last trip of the day, the algorithm does not have a "next trip" to deduce their destination. In these cases, reconstructing the flow requires crossing the card's temporal history with the physical route of the bus modeled in the GTFS. By merging the high-volume validation records with the spatial density of the GTFS through multidimensional join operations, it becomes possible to geographically map the passenger flow and locate hidden bottlenecks within the network.

References

- [1] Directorio de Transporte Público Metropolitano (DTPM), "Matrices de Viaje y Datos GTFS del Sistema de Transporte," *Portal de Datos y Servicios DTPM*. [Online]. Available: <https://www.dtpm.cl/>.
- [2] Ministerio de Transportes y Telecomunicaciones (MTT), "Portal de Datos Abiertos del Estado," *Datos.gob.cl*. [Online]. Available: <https://datos.gob.cl/>.